

EE150 Signal and System

Homework 4

Note:

- Please provide enough calculation process to get full marks.
- Please submit your homework to Gradescope in PDF version.
- It's highly recommended to write every exercise on a single sheet of page.
- Late submissions will have points deducted according to the penalty policy.
- Please use English only to complete the assignment, solutions in Chinese are deducted according to the penalty policy.
- Plagiarizer will get zero points.
- The full score of this assignment is 100 points.

Exercise 1. (12pt)

Using Fourier transform analysis, determine the Fourier transform of the following signals:

(a) $e^{-2|t-1|}$

(b) $\frac{d}{dt}[u(-2-t) + u(t-2)]$

(c) $1 + \cos\left(6\pi t + \frac{\pi}{8}\right)$

Exercise 2. (12pt)

Given that the Fourier transform of $x(t)$ is $X(j\omega)$ and $X(0) = 2$, express the Fourier transforms of the following signals in terms of $X(j\omega)$ using the property of Fourier transform.

(a) $x_1(t) = 2x(t-1) + x(2-t)$

(b) $x_2(t) = x(4t+2)$

(c) $x_3(t) = \int_{-\infty}^t x(2\tau-1) d\tau$

Exercise 3. (15pt)

Consider a signal $x(t)$ with Fourier transform $X(j\omega)$, and suppose the following conditions are given:

1. $x(t)$ is real-valued and nonnegative.
2. $\mathcal{F}^{-1}\{(1 + j\omega)X(j\omega)\} = Ae^{-2t}u(t)$, where A is independent of t .
3. $\int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega = 2\pi$.

Find a closed-form expression for $x(t)$.

Exercise 4. (10pt)

Consider a stable LTI system with frequency response

$$H(j\omega) = \frac{1}{j\omega + 3}.$$

For a particular input $x(t)$ this system is observed to produce the output

$$y(t) = e^{-2t}u(t) - e^{-3t}u(t).$$

Determine the frequency response $X(j\omega)$ and $x(t)$.

Exercise 5. (15pt)

Given

$$x(t) = e^{-|t|}$$

(a): Calculate the Fourier transform of $x(t)$.

(b): Using the appropriate Fourier transform property, find the inverse Fourier transform $y(t)$ of $Y(j\omega) = \frac{-4j\omega}{(1+\omega^2)^2}$.

(c): Using the result from part (b), together with the duality property, find the Fourier transform of

$$\frac{4t}{(1+t^2)^2}.$$

Exercise 6. (21pt)

Let

$$y(t) = x(t) \cos^2 t$$

$$g(t) = y(t) * \frac{\sin t}{\pi t}, \text{ (* means convolution operator)}$$

Assume that $x(t)$ is a real-valued signal with $X(j\omega)$ shown below

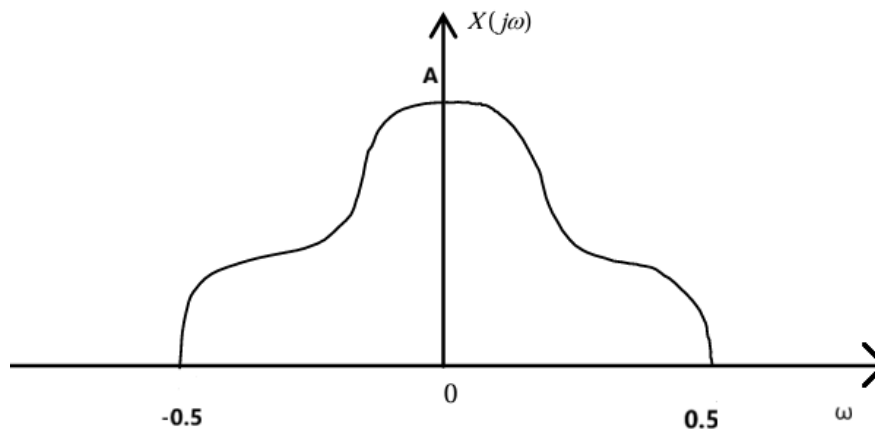


Figure 1: $X(j\omega)$

Suppose $X(j\omega) = A$ when $\omega = 0$.

- (a): Find the Fourier transform $Y(j\omega)$ of $y(t)$ and sketch $Y(j\omega)$
- (b): Find the Fourier transform $G(j\omega)$ of $g(t)$ and sketch $G(j\omega)$

Exercise 7. (15pt)

The input and output of a stable LTI system are characterized by the following differential equation:

$$\frac{d^2y(t)}{dt^2} + 6\frac{dy(t)}{dt} + 8y(t) = 2x(t)$$

(a) Find the unit impulse response of this system by solving the equation of $X(j\omega)$ and $Y(j\omega)$.

(b) If $x(t) = \delta(t) - e^{-3t}u(t)$ is the input to the system and the response of this system is $y(t)$. The Fourier transform of $y(t)$ is $Y(j\omega)$. Find $Y(j\omega)$ and $y(t)$.